Auditing Annotation Support and Evaluation Sensitivity in Aerial Vehicle Detection

Xinling Wen, Jiahao Zhang, Wenqi Liu, Yangxu Ren, and Yu Chen

***Abstract*—Aerial detection artifacts are frequently compared under an implicit assumption of comparable evaluation support. We formalize an audit protocol with three quantities: annotation-support shift, an evaluation-policy band, and candidate-pool rank stability. The audit covers VisDrone validation, a provenance-checked local UAVDT conversion, AI-TOD validation, and DOTA-v2 validation (structural-only), totaling 141,347 vehicle boxes. Seven frozen prediction artifacts are replayed over 9,408 prespecified greedy-rule records; raw prediction coverage is determined before class filtering, so a covered image with no vehicle prediction remains an empty prediction. Fixed 24-pixel filtering retains 73.3%, 95.1%, and 9.7% of vehicle boxes in VisDrone, local UAVDT, and AI-TOD, respectively. AP$_{50}$ bands reach .097 and F_1 bands reach .330. In the two multi-candidate source pools, deterministic AP$_{50}$ winners are unchanged in 12 headline source–rule–policy checks (10 nonduplicate policy outcomes); greedy and threshold-first Hungarian matching agree on winners in 24 checks (20 nonduplicate outcomes). In paired 1,000-replicate comparisons of the leading and runner-up candidates, every VisDrone AP$_{50}$/F_1 95% interval favors SAHI, whereas every local-UAVDT sequence-proxy interval contains zero. The audit characterizes declared artifacts and frozen configuration pools; it neither estimates annotation-error rates nor establishes a detector leaderboard.**

***Index Terms*—Aerial detection, annotation support, evaluation sensitivity, benchmark auditing, vehicle detection.**

I. Introduction

Results in aerial detection depend jointly on prediction artifacts and the annotation support to which they are compared. Platform, resolution, ontology, ignored regions, small-object handling, and matching conventions can change the target of a reported metric. Existing aerial benchmarks are valuable but deliberately heterogeneous [1], [2], [3], [4], [5], [6]; label-space heterogeneity and dataset-specific bias further complicate comparison [7], [8]. Recent GRSL studies span efficient high-resolution image modeling and explicit analysis of validation bias [9], [10]; neither addresses annotation-support/evaluator interactions in aerial detection. Error taxonomies and label-error studies answer complementary questions [11], [12], [13]. They do not jointly quantify how declared annotation support, an evaluation policy, and a fixed candidate pool affect an artifact-scoped conclusion.

Small-object treatments are especially consequential when pixel support is limited [6], while label-issue analyses inspect a different failure mode [14]. Our protocol does not replace either line of work: it records the target being scored before making a cross-artifact comparison.

We therefore define a three-part audit: (i) *annotation-support shift*, the change in retained valid boxes under a declared support rule; (ii) an *evaluation-policy band*, $\max_r m(r) - \min_r m(r)$ over prespecified rules r; and (iii) *candidate-pool rank stability*, agreement of the winner within a declared multi-candidate source pool. This is an evaluation-audit protocol, not a new detector or a claim about label correctness.

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TABLE I
Declared artifact support and raw-prediction coverage. A covered image with no retained vehicle-class prediction is evaluated as empty.

Source	Eval. images	Vehicle+ images	Covered images	Boxes	Median px side	Median norm. side
VisDrone	548	519	548	17,040	41.0	.0412
Local UAVDT	305	305	305	11,054	33.0	.0278
AI-TOD	2,804	1,443	1,869	59,904	14.0	.0175
DOTA-v2	5,108	2,581	–	53,349	18.0	.0352

II. Artifacts, Coverage, and Protocol

We use VisDrone2019-DET validation, a local COCO-style UAVDT conversion, AI-TOD validation, and DOTA-v2 patched validation [1], [3], [6], [15]. AI-TOD's available crop reconstruction is xView-derived [5]. The vehicle mapping is VisDrone {car, van, truck, bus}; UAVDT {car, truck, bus}; AI-TOD {vehicle}; and DOTA-v2 {small-vehicle, large-vehicle}. VisDrone source ignored regions are retained as ignores; truncation/occlusion attributes are not filtered. UAVDT uses its local COCO $xywh$ conversion and is therefore conversion-conditioned. DOTA quadrilaterals are converted to axis-aligned boxes for structural scale only; it has no frozen prediction artifact. The patched DOTA validation tar-stream artifact and all conversion details are recorded in the provenance manifest.

AI-TOD coverage is incomplete: the 1,869 covered images contain 16,900 vehicle boxes (9.0/image; median side 12 px), versus 43,004 boxes in 935 noncovered images (46.0/image; median side 16 px). Thus AI-TOD supports only a single-candidate, coverage-conditioned sensitivity analysis and no rank-stability claim. VisDrone and local UAVDT have complete source-wise common coverage (548/548 and 305/305).

Coverage is defined by the presence of an image name in the *raw* prediction artifact, before the vehicle-class subset is taken. This distinguishes an executed image with zero retained vehicle predictions from a missing cache record. After filtering, the former contributes all valid GT as false negatives and no retained predictions; it is never silently removed from F1 or AP. Candidate-specific coverage is retained as a supplementary manifest, while all within-source candidate comparisons in this paper use the source-wise common coverage in Table I. The pronounced AI-TOD covered/noncovered contrast above is reported as selection evidence, not corrected by extrapolation.

For a box (w, h) in image (W, H), we use absolute support $s_a = \max(w, h)$ and normalized support $s_n = \max(w/W, h/H)$. The full grid uses $s_a \in \{16, 20, 24, 28, 32, 40, 48, 64\}$ px, $s_n \in \{.005, .0075, .010, .015, .020, .030\}$, confidence $\in \{.10, .15, .20, .25, .30, .35, .40, .50\}$, and IoU $\in \{.25, .30, .40, .50\}$. These values were *prespecified* in the versioned rule manifest before headline computation. Headline rules use 24 px and .015 because they are registered interior grid values and expose the observed absolute/normalized support contrast.

The full surface contains $7 \times (8 \times 4 \times 8 \times 3 + 8 \times 4 \times 6 \times 3) = 9{,}408$ greedy records. The headline AP surface uses all seven candidates, two scale rules, and three policies (42 records); AP has no confidence dimension because it integrates score order. The matching control uses the same two rules, three policies, confidence .25, and IoU .25/.50 (84 records). These different counts identify evaluated records rather than independent rank tests. For the pairwise diagnostics, $\mathcal{R}_h^{AP} = \{(a, 24\,\mathrm{px}), (n, .015)\} \times \Pi$ contains six nominal scale–policy settings. $\mathcal{R}_h^{F_1} = \{(a, 24\,\mathrm{px}), (n, .015)\} \times \Pi$ fixes $c = .25$ and $\iota = .25$, and likewise contains six settings. The separate matching control is $\mathcal{R}_h^{match} = \{(a, 24\,\mathrm{px}), (n, .015)\} \times \Pi \times \{.25, .50\}$ at $c = .25$ (84 candidate records). Exact metric ties are counted as rank instability; candidate or file order is never used as a tie-break.

A. Policy and matching contract

The authoritative primary-grid policy set is $\Pi = \{I, O, N\}$, where I=include-all, O=exclude/source-ignore-off, and N=exclude/source-ignore-on. It alone defines the 9,408 greedy records, 42 AP records, 84 matching records, and every policy band in Fig. 2. *Include-all* retains all valid mapped boxes. *Exclude/source-ignore-off* removes below-threshold valid boxes while leaving predictions eligible as false positives. *Exclude/source-ignore-on* makes the same removal but suppresses predictions overlapping source-provided ignores at .5 IoU. Removed valid boxes are not source ignores in either primary-grid exclusion policy. The fourth rule, *exclude-and-ignore-removed*, is an *ancillary out-of-grid diagnostic*: removed valid boxes and source ignores suppress overlapping predictions, but removed boxes are absent from recall. It is reported separately only to isolate support removal from a penalty for detecting removed objects.

Greedy matching orders predictions by score and assigns the best currently unmatched eligible GT. Hungarian matching is threshold-first lexicographic: valid edges satisfy the declared IoU; the assignment first maximizes their cardinality and then their total IoU, with dummy nodes permitting unmatched objects. Fixed-operating-point performance is global micro-$F_1 = 2TP/(2TP + FP + FN)$ after pooling the evaluated images. AP uses COCO bbox evaluation [16] with complete score ordering, common category 1, the same image set, standard area ranges and `maxDets`=100; it is not crossed with a fixed confidence threshold. Under an exclusion policy, removed valid boxes are absent from AP GT; source-ignore-on suppresses its registered overlapping predictions, and removed boxes are not converted to COCO crowd/ignore annotations. The one-to-one overlap contract follows the standard detection-evaluation framing [17]. Raw caches are retained at their recorded scores and hashes in the candidate manifest.

For source s, write a full rule as $r = (d, \tau, \pi, c, \iota)$: scale domain $d \in \{a, n\}$, threshold τ, small-object policy π, confidence c, and IoU ι. The support shift is $1 - |G_s(d, \tau)|/|G_s|$. At fixed (d, τ, c, ι), the reported *policy band* is $B_m^\pi = \max_{\pi \in \Pi_s} m(d, \tau, \pi, c, \iota) - \min_{\pi \in \Pi_s} m(d, \tau, \pi, c, \iota)$. We do not report a Cartesian full-rule band as a headline quantity. A rank check is defined only when at least two candidates share C_s; its deterministic stability is the fraction of such fixed settings for which the same candidate maximizes m. These definitions prevent a single-candidate result from being counted as rank stability.

For a finite headline rule set $\mathcal{R}_h$, define the candidate-level *rule band* $B_k(\mathcal{R}_h) = \max_{r \in \mathcal{R}_h} m_k(r) - \min_{r \in \mathcal{R}_h} m_k(r)$. This is distinct from the fixed-(d, τ, c, ι) policy band B_m^π above. For a reference rule $r_0 \in \mathcal{R}_h$ and its winner $k^\star$, define $\Delta_j(r) = m_{k^\star}(r) - m_j(r)$ and $\Gamma_j(\mathcal{R}_h) = \max_{r \in \mathcal{R}_h} |\Delta_j(r) - \Delta_j(r_0)|$. Unlike an individual policy band, $\Gamma_j(\mathcal{R}_h)$ measures only rule-induced movement of a pairwise gap. **Lemma 1 (resize invariance).** Under isotropic resizing by $\alpha > 0$, absolute support becomes αs_a, whereas normalized support remains s_n; hence normalized-side thresholds are resize-invariant. **Proposition 1 (policy-robust winner).** If $\Delta_j(r_0) > \Gamma_j(\mathcal{R}_h)$ for every competitor j, then $k^\star$ remains the winner for every $r \in \mathcal{R}_h$. Indeed, $\Delta_j(r) \geq \Delta_j(r_0) - \Gamma_j(\mathcal{R}_h) > 0$. Moreover, $\Gamma_j(\mathcal{R}_h) \leq B_{k^\star}(\mathcal{R}_h) + B_j(\mathcal{R}_h)$, which supplies a conservative sufficient condition. Formal proofs, a non-common-coverage non-identifiability result, and an F_1 perturbation bound appear in the supplementary material.

For uncertainty, we resample common evaluation units jointly for the first and second candidates: images for VisDrone and filename-derived sequence proxies for local UAVDT. The UAVDT proxy is the substring before the first underscore in the image name; it yields 93 units over 305 images (min/median/max unit sizes 1/1/104). Each of 1,000 paired replicates (seed 20260712) recomputes global F_1 from resampled TP/FP/FN totals and recomputes the score-ordered one-category COCO AP_{50} slice (IoU .5, `maxDets`=100), preserving pairing; intervals are percentile 2.5/97.5% intervals. Setting-level point differences, intervals, and winner probabilities are released in the supplementary repository. This cluster proxy is a sensitivity device, not a claim that filename prefixes recover official UAVDT sequences. We consequently report its interval as qualification rather than converting it to a detector-wide significance claim.

TABLE II
FROZEN CANDIDATE POOL AND INCLUDE-ALL REFERENCE AP_{50} POINT ESTIMATES AT MAXDETS=100. ALL RAW CACHES AVAILABLE AT AUDIT FREEZE WERE INCLUDED; THIS IS A CONFIGURATION POOL, NOT A DETECTOR-FAMILY SAMPLE.

Source	Frozen pool	Reference AP_{50}@100	Role
VisDrone	baseline640; SAHI640	SAHI .718 / base .364	two-variant pool
Local UAVDT	baseline; tiling; YOLO11n; YOLO11m	tiling .249 / base .215	four-variant pool
AI-TOD	baseline640	baseline .234	single-candidate sensitivity
DOTA-v2	none	–	structural-only

III. RESULTS

Figure 1 makes the denominator and target-set shift visible together. At 24 px, VisDrone, local UAVDT, and AI-TOD retain 73.3%, 95.1%, and 9.7% of boxes; at .015 they retain 89.8%, 97.4%, and 81.0%. The unequal absolute-rule curves show why a common pixel cutoff does not define a common evaluation target. These distributions are artifact properties, not evidence that one source is more correct. The normalized-support Wasserstein matrix is retained as Supplementary Fig. S1 rather than used as a headline result.

The 24-px excluded fractions are 26.7% [24.4,29.2], 4.9% [0.4,7.0], and 90.3% [89.3,91.3] for VisDrone, local UAVDT, and AI-TOD (cluster-bootstrap 95% intervals). Local UAVDT intervals use filename-derived sequence proxies; their resampled center need not equal the pooled box-weighted point estimate. At confidence/IoU .25, the corresponding primary F_1 policy bands are .093, .015, and .330; normalized-rule bands are .061, .130, and .136. Under exclude-and-ignore-removed, AI-TOD F_1 is .130 (24 px) and .317 (.015), versus .065 and .260 under exclusion without protection. Thus its large bands reflect both extreme support removal and the treatment of predictions on removed targets.

Figure 2 shows that the headline thresholds are not isolated anomalies. AI-TOD has broad high-sensitivity regions under absolute filtering; local UAVDT becomes strongly sensitive at the larger normalized thresholds; and VisDrone varies more gradually. Across the complete displayed surfaces, maximum bands are .429/.235 for VisDrone, .512/.493 for local UAVDT, and .464/.368 for AI-TOD under absolute/normalized support, respectively. These are candidate-specific sensitivity surfaces, not cross-source performance comparisons.

The AP headline has 42 candidate records at the registered `maxDets`=100; Table II gives the include-all reference AP_{50} point values. For the two multi-candidate source pools, the AP_{50} winner is unchanged in all 12 nominal source–rule–policy checks: SAHI for VisDrone and tiling for local UAVDT. Because the two source-ignore policies are identical for local UAVDT, these are 10 distinct policy outcomes. AP_{50} bands range from .001–.097 in local UAVDT and .023–.046 in VisDrone; AI-TOD has one candidate and is explicitly excluded from this rank statement. The paired AP_{50} bootstrap qualifies this deterministic finding. Across the six VisDrone settings, the individual 95% intervals have lower endpoints .339–.362 and upper endpoints .366–.387, so each excludes zero. Across the four local-UAVDT settings, every individual sequence-proxy interval contains zero (lower endpoints $-.310$–$-.296$, upper endpoints .071–.111; winner probability .471–.604). *Detection-cap sensitivity is material.* After source-ignore handling, `maxDets`=100 truncates predictions in 406–490 of 548 VisDrone images and 272–292 of 305 local-UAVDT images across the primary policies. Focused `maxDets`=300 and 2000 controls retain the same pairwise winner in all 10 nonduplicate multi-candidate outcomes; at 2000 no retained prediction is truncated (largest per-image count 1,066), but pairwise AP_{50} gaps relative to 100 shift by $[-.041, .048]$. Per-setting point estimates and truncation counts are released with the supplementary repository.

For 84 greedy/Hungarian F_1 records (seven candidates, two rules, three policies, two IoUs), the 24 multi-candidate checks have identical winners; 20 remain after collapsing duplicate source-ignore outcomes. The maximum absolute F_1 difference is .00546, so matching is not metric-identical (Supplementary Fig. S2). Figure 3 provides the necessary uncertainty qualification: all VisDrone AP_{50}/F_1 intervals are positive, whereas every local-UAVDT sequence-proxy interval contains zero. Hence the local UAVDT deterministic winner should not be read as statistically established under that resampling unit.

A. Differential policy diagnostics

Figure 3(a) separates deterministic and resampling-qualified conclusions. All four points satisfy the certificate on their declared headline rule sets. VisDrone AP_{50} has a reference SAHI–baseline margin of .354 and a differential radius of .021; the observed minimum margin remains .354 because the pairwise gap increases rather than decreases under the relevant rules. In contrast, local UAVDT has an AP_{50} reference margin of .034 and $\Gamma = .033$, leaving a smallest observed tiling–baseline margin of only .0014. Its deterministic certificate therefore holds, while its paired sequence-proxy bootstrap does not establish an advantage. For F_1, the local-UAVDT differential radius is only .00007 and the minimum margin is .191. Thus a large individual policy band is neither necessary nor sufficient for rank reversal; the relevant object is movement of the pairwise margin.

This distinction also explains a common-mode effect. If a rule lowers two candidates by the same amount, both individual bands can exceed their reference gap while $\Delta_j(r)$ is unchanged. Conversely, even small individual bands can reverse a close ranking if their shifts are anti-aligned. The conservative corollary based on $B_{k^\star}(\mathcal{R}_h) + B_j(\mathcal{R}_h)$ cannot recognize this correlation, whereas $\Gamma_j(\mathcal{R}_h)$ does. We therefore report individual bands as a sensitivity description and pairwise differential radii as a deterministic rank diagnostic. Neither replaces uncertainty quantification: a positive minimum margin is a statement about the declared artifacts and rules, while bootstrap winner probability characterizes a specified resampling unit.

B. Why common coverage is required

The common-coverage requirement is not merely a bookkeeping convention. If candidates A and B are observed on different image subsets C_A and C_B, their ordering on the full source cannot be identified from subset metrics alone. Holding all observed predictions fixed, assigning favorable unobserved

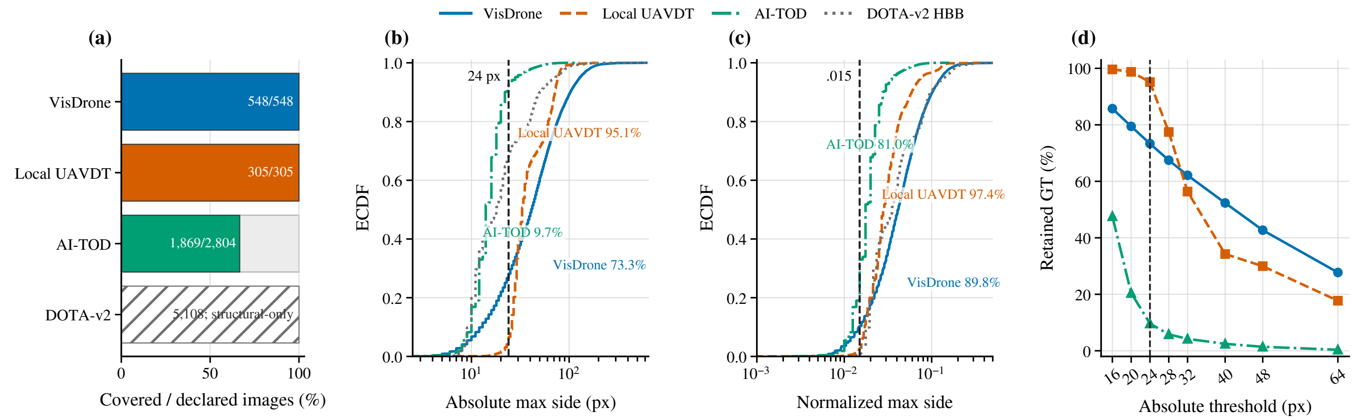

Fig. 1. Dataset support and coverage structure. (a) Source-wise common raw-prediction coverage; DOTA-v2 has no prediction artifact and is structural-only. (b),(c) Equal-box-weight ECDFs of absolute and normalized horizontal-box maximum side; labels give the percentage retained at the 24-px and .015 headline thresholds. DOTA-v2 uses an axis-aligned HBB proxy derived from quadrilaterals. (d) Empirical retained-GT curves over the complete absolute threshold grid for the three evaluated sources.

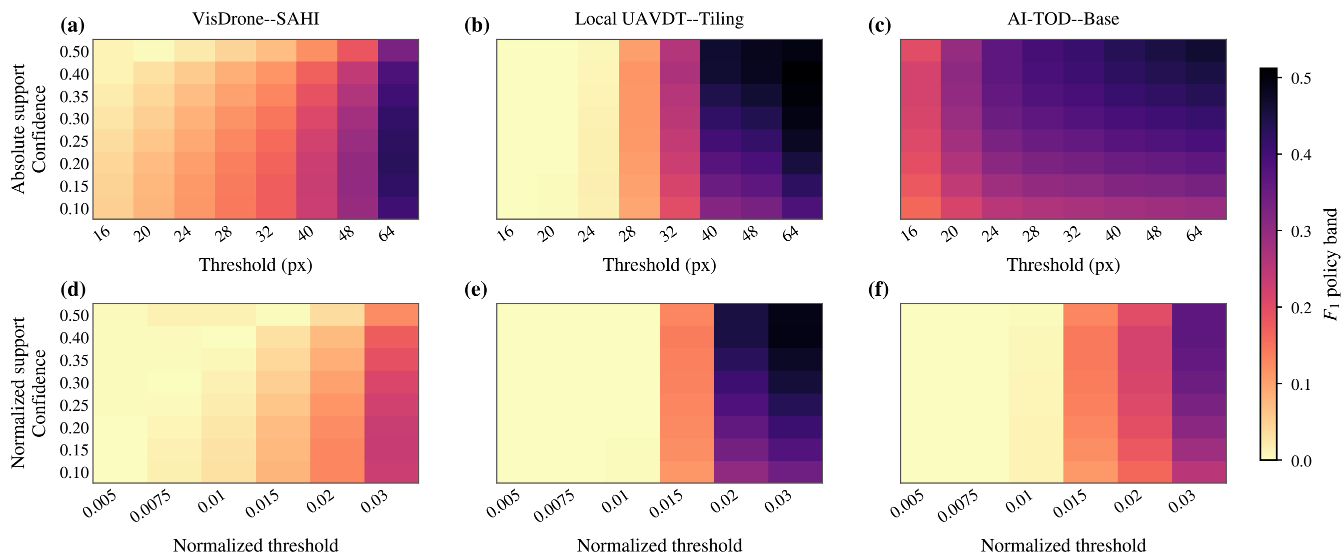

Fig. 2. Full F_1 evaluation-sensitivity surfaces at IoU .25. Columns use the declared VisDrone–SAHI, local-UAVDT–tiling, and coverage-conditioned AI-TOD–baseline artifacts; rows use absolute and normalized support. Each cell is $\max_{\pi\in\Pi} F_1 - \min_{\pi\in\Pi} F_1$ over the three primary policies at a fixed support threshold and confidence. All panels share one color scale.

outcomes to A produces one compatible full-source ordering, while assigning unfavorable outcomes produces the opposite ordering. Any full-population rank statement therefore needs either a shared denominator or explicit assumptions about unobserved images. This is why AI-TOD, despite its useful structural and policy-sensitivity evidence, has no rank-stability result; and why all reported VisDrone and local-UAVDT comparisons include covered empty predictions rather than silently intersecting after class filtering. A formal constructive proof and an F_1 count-perturbation bound are given in the supplement.

IV. Discussion, Scope, and Reproducibility

The contribution is a reporting protocol: disclose artifact lineage and support, report a policy band under explicit treatment of removed objects, and restrict rank claims to common-coverage multi-candidate pools with uncertainty qualification. It does not make a detector-wide leaderboard, estimate label-error rates, or infer physical traffic density. Count MAE comparisons use the same common image set within each source: SAHI versus VisDrone baseline is 9.0 versus 62.5, and local UAVDT tiling versus baseline is 16.8 versus 89.4, both on complete source-wise common coverage.

The frozen pool contains all seven available cached raw-prediction artifacts at audit freeze: two VisDrone variants (baseline and SAHI [18]), four local UAVDT variants (baseline, tiling, YOLO11n, YOLO11m), and one AI-TOD baseline. Its asymmetry is a boundary, not evidence of broad detector-family behavior. Code, frozen manifests, intermediate evaluation records, figure-source data, and SHA-256 checksums are available at https://anonymous.4open.science/r/aerial-evaluation-audit-5565/README.md. DOTA remains structural-only. These records make it possible to distinguish missing prediction coverage from an evaluated empty predic-

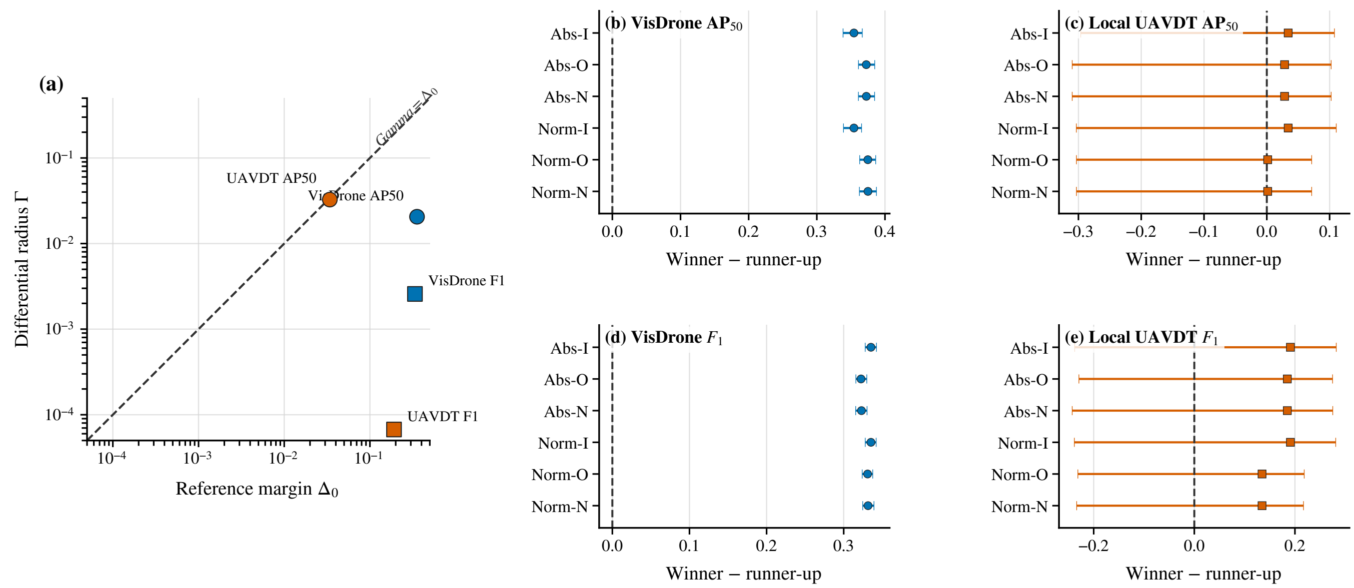

Fig. 3. Rank robustness and uncertainty. (a) Deterministic geometry on logarithmic axes: all declared pairwise points satisfy $\Gamma < \Delta_0$ and therefore lie below the reversal boundary. (b)–(e) Paired-bootstrap winner–runner-up differences and separate 95% intervals for six nominal headline scale–policy settings at AP_{50} or F_1 (confidence/IoU .25 for F_1). I/O/N denote include-all, exclude/source-ignore-off, and exclude/source-ignore-on. Local-UAVDT AP_{50} N rows duplicate O because the registered source has no effective source-ignore distinction.

tion.

V. Conclusion

Cross-source aerial evaluation should report support shift, policy bands, and rank stability separately. In these declared artifacts, a 24-px rule retains radically different support across sources, while normalized filtering mitigates but does not eliminate the disparity. Rules can substantially change AP_{50}/F_1, and deterministic winners can remain unchanged; however, bootstrap evidence shows that this does not by itself establish a stable local UAVDT advantage. The practical recommendation is to publish the image-coverage denominator, removed-object treatment, exact matching contract, and uncertainty-qualified candidate-pool result alongside every cross-source comparison.

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